

Advanced (Habanero)

- Q1.** What is the solution to the system $x + y = 2$ and $x - y = 0$?
(A) $(-1, 1)$
(B) $(1, 1)$
(C) $(0, 0)$
(D) $(2, 0)$
(E) $(0, 2)$
- Q2.** The system $2x + 3y = 7$ and $x - 2y = -3$ has:
(A) One solution
(B) No solutions
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions
- Q3.** To find the solution to a system by graphing, what must be done?
(A) Solve for x
(B) Solve for y
(C) Graph each equation and find the intersection point
(D) Add the equations
(E) Subtract the equations
- Q4.** What is the x -coordinate of the solution to the system $y = 2x - 1$ and $y = -x + 3$?
(A) -2
(B) 1
(C) 0
(D) 2
(E) 4
- Q5.** The system $x + 2y = 4$ and $3x - 2y = 2$ is an example of:
(A) Consistent and independent
(B) Consistent and dependent
(C) Inconsistent
(D) Dependent
(E) Independent
- Q6.** The system $y = x + 2$ and $y = x - 2$ has:
(A) One solution
(B) No solutions
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions
- Q7.** What is the y -coordinate of the solution to the system $x - y = 2$ and $2x + 2y = 4$?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q8.** The system $y = x^2$ and $y = 4$ has:
(A) One solution
(B) No solutions
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions

Open Ended Questions (FRQ)

- Q9.** Solve the system $2x + 3y = 7$ and $x - 2y = -3$ by graphing. Show your work and identify the solution.
- Q10.** Consider the system $y = 2x - 1$ and $y = -x + 3$. Graph both equations on the same coordinate plane and find the intersection point. What are the coordinates of this point?

Chef's Tips

- Emphasize the importance of accurately graphing the equations
- Remind students to check for any calculation errors
- Encourage students to verify their solutions by plugging them back into the original equations